Career Launcher MBA

QA Arithmetic – 07

Mixtures and Solutions; Replacement

Question 1

In what ratio should 30% Phenyl Alcohol should be mixed with 60% of Phenyl Alcohol solution so that the resultant solution has 50% Phenyl Alcohol in it?

(1) 1 : 2

(2) 2 : 1

(3) 1 : 3

(4) 3 : 1

Question 2

Two liquids A and B are mixed in the ratio of 3 : 2 respectively and the mixture is sold at Rs. 11 per litre at 10% profit. If the first liquid costs Rs. 2 per litre more than the second, the cost (rupees per litre) of first liquid is

(1) 10.8

(2) 8.8

(3) 18.8

(4) 5

Question 3

A solution of 20 L of wine and water contains 80% wine. How much water (in L) must be added to this solution in order to make it a 40% wine solution?

Question 4

There are two alloys of copper and aluminium in the ratio 7 : 2 and 7 : 11 respectively. If equal quantities of the two alloys are melted to form a third alloy, then find the ratio of copper to aluminium in newly fomed alloy.

(1) 3 : 4

(2) 5 : 3

(3) 7 : 3

(4) 7 : 5

Question 5

After the addition of 35 liters of water to a can of diluted milk, the concentration of milk in the can becomes 30%. Now, further 40 liters of water is added to the can and the concentration of milk in the can gets reduced by 10 percentage points. How many more liters of water must be added to the can now such that the concentration of milk in the can becomes 8%?

Question 6

A 20% ethanol solution is mixed with another ethanol solution, say, S of unknown concentration in the proportion 1:3 by volume. This mixture is then mixed with an equal volume of 20% ethanol solution. If the resultant mixture is a 31.25% ethanol solution, then the unknown concentration of S is

(1) 50%

(2) 48%

(3) 55%

(4) 52%

Question 7

Beaker A and beaker B contains methanol, ethanol and phenyl in the ratio 1 : 3 : 2 and 2 : 1 : 5 respectively. Some parts of the solutions from beaker A and beaker B are thoroughly mixed and put into another beaker C. Which of the following cannot be the ratio of methanol, phenyl and ethanol in beaker C?

(1) 10 : 23 : 15

(2) 7 : 15 : 16

(3) 6 : 13 : 13

(4) 9 : 20 : 18

Question 8

If a certain weight of an alloy of silver and copper is mixed with 3 kg of pure silver, the resulting alloy will have 90% silver by weight. If the same weight of the initial alloy is mixed with 2 kg of another alloy which has 90% silver by weight, the resulting alloy will have 84% silver by weight. Then, the weight of the initial alloy, in kg, is

(1) 3

(2) 2.5

(3) 3.5

(4) 4

Question 9

Product M is produced by mixing chemical X and chemical Y in the ratio of 5 : 4. Chemical X is prepared by mixing two raw materials, A and B, in the ratio of 1 : 3. Chemical Y is prepared by mixing raw materials, B and C, in the ratio of 2 : 1. Then the final mixture is prepared by mixing 864 units of product M with water. If the concentration of the raw material B in the final mixture is 50%, how much water was added to product M?

(1) 328 units

(2) 368 units

(3) 392 units

(4) 616 units

Question 10

The volume of a sugar solution in three different vessels form an arithmetic progression with a common difference of 5 kg. All the three sugar solutions have the same percentage concentration of sugar. The difference between the numerical values of the concentration (in percentage) of sugar in the sugar solution and the volume of sugar solution in the vessel having the least quantity is 8. The total quantity of sugar in the three vessels is 5.4 litres. What are the quantities of sugar solution in the three vessels?

(1) 5 kg, 10 kg and 15 kg

(2) 6 kg, 11 kg and 16 kg

(3) 7 kg, 12 kg and 17 kg

(4) 8 kg, 13 kg and 18 kg

Question 11

Three milkmen Ramu, Kamu and Namu had 5L, 10L and 15L of milk-water solution respectively. In the solution that Kamu and Namu had, the ratio of milk and water were 2 : 3 and 1 : 4 respectively. All of them sat together and mixed their respective solutions together. They sold the combined mixture at Rs. 20 per litre and Ramu got Rs. 2.5 for every litre of mixture sold. What was the milk-water ratio in the solution that Ramu had with him? (Assume water comes free of cost and they share the revenue earned in the ratio of milk content)

(1) 1 : 4

(2) 2 : 3

(3) 1 : 5

(4) 3 : 5

Question 12

Two alloys of iron have different percentages of iron in them. The first of them weighs 6 kg and the second one weighs 12 kg. One piece each, of equal weight, was cut off from both the alloys. The piece from the first was mixed with the leftover part of the second alloy and the piece from the second with the leftover part of the first alloy. As a result the percentage of iron became the same in both the alloys. What was the weight of each cut out pieces?

(1) 4 kg

(2) 3.5 kg

(3) 3 kg

(4) None of these

Question 13

In a mixture of 120 L, the ratio of milk and water is 2 : 3. If 30 L of mixture is replaced by 18 L of milk and 12 L of water, then new ratio of milk to water is

(1) 7 : 9

(2) 9 : 11

(3) 2 : 3

(4) 11 : 9

Question 14

A can contains a solution of two liquids A and B in the ratio of 7 : 5. When 9 L of solution is drawn off from the solution and the can is filled with B, the proportion of A and B becomes 7 : 9. How many litres of liquid A was there in the can initially?

Question 15

A beaker contains milk and water in the ratio 5 : 3. If a person removes 6 L of this mixture and replaces it with pure water, the ratio of milk to water becomes 1 : 1. What is the amount of milk present in beaker initially?

(1) 18.75 L

(2) 16.25 L

(3) 20 L

(4) 18 L

Question 16

One-litre can is full with a mixture of water and alcohol, and alcohol content is 20% of the total mixture. If 10% of this liquid is taken out and pure alcohol is added to fill the can, what will be the percentage of alcohol in the new mixture?

Question 17

Two jars contain milk and water in the ratio 5 : 4 and 2 : 1 respectively. What volume should be taken out from the first jar and second jar so as to fill up a third 30 L jar with milk to water in the ratio 1 : 1?

(1) 7.5 L

(2) 15 L

(3) 22.5 L

(4) Impossible situation

Question 18

A vessel contains 180 L wine. 60 L is taken out of the vessel every day and an equal volume of water is put in it. What volume of wine remains in the vessel at the end of 3 days?

(1) 160/3 L

(2) 20/3 L

(3) 80 L

(4) 35 L

Question 19

Some amount of acid is taken out of a 54 L vessel full of acid and then an equal amount of water is added to it. The process is repeated once more. As a result, the vessel now contains 24 L pure acid. How much of the acid was taken out initially (in L)?

Question 20

A bottle contains 50 liters of milk. From this bottle 5 liters of milk was taken out and replaced by water. This process was repeated further for three times. How much milk is now contained in the bottle?

(1) 32.8 liters

(2) 34.4 liters

(3) 36.8 liters

(4) 46.5 liters

Question 21

A cylindrical container is having the top surface of a liquid at a level of 4/7 of its full level. From this, if x litres is removed, the container is one-fourth full. Finally, 35 L is added to the container from this position, making it half full. Find the value of x and the capacity of the container respectively.

(1) 45 L and 210 L

(2) 25 L and 140 L

(3) 45 L and 140 L

(4) 40 L and 210 L

Question 22

An alloy is prepared by mixing three metals A, B and C in the proportion 3 : 4 : 7 by volume. Weights of the same volume of the metals A, B and C are in the ratio 5 : 2 : 6. In 130 kg of the alloy, the weight, in kg, of the metal C is

(1) 84

(2) 70

(3) 48

(4) 96

Question 23

A solution, of volume 40 litres, has dye and water in the proportion 2 : 3. Water is added to the solution to change this proportion to 2 : 5. If one-fourths of this diluted solution is taken out, how many litres of dye must be added to the remaining solution to bring the proportion back to 2 : 3?

Question 24

Pipe X pours a mixture of acid and water, and pipe Y pours pure water into a bucket. After 1 hour, the bucket got filled and the concentration of acid in the bucket was noted to be 8%. If pipe Y was closed after 30 minutes and pipe X continued to pour the mixture, concentration of acid in the bucket after 1 hour would have been 10%. What is the ratio of acid to water in the mixture coming out of pipe X?

(1) 13 : 2

(2) 2 : 15

(3) 3 : 20

(4) 2 : 13

Question 25

From a container filled with milk, 9 litres of milk are drawn and replaced with water. Next, from the same container, 9 litres are drawn and again replaced with water. If the volumes of milk and water in the container are now in the ratio of 16 : 9, then the capacity of the container, in litres, is

Thanks!